Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Predictive Modeling, Machine Learning, Generative & Agentic AI, Dimensional Data Modeling, Star/Snowflake Schemas, Slowly Changing Dimensions (SCD Type 1/2), Time-Series Forecasting, Automated Data Quality Audits, ETL Pipeline Optimization, Root Cause Analysis, Capacity Planning, Statistical Techniques, Scalable Data Architectures

Programming Languages & Tools: Python, PySpark, Apache Spark, SQL, XGBoost, Scikit-Learn, SARIMA, Azure Data Factory, Azure Synapse, Databricks, AWS (S3, Redshift), Snowflake, Power BI, Tableau, MicroStrategy, Airflow, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data & AI Solutions Engineer — **May 2022 – Present**

- **Architected scalable predictive modeling and AI pipelines** utilizing XGBoost, Random Forests, and Scikit-Learn to drive customer behavior modeling and operational reliability by 20%, collaborating closely with cross-functional stakeholders.
- **Engineered advanced time-series forecasting models (SARIMA)** to optimize workforce capacity planning and predict customer truck rolls, reducing service response times by 20% across large-scale distributed datasets.
- **Deployed Agentic AI frameworks and automated QA testing** via SpectrumIQ and prompt engineering (KIRO IDE), cutting manual testing turnaround by 50% and accelerating end-to-end data engineering deliverables.
- **Designed robust dimensional data models and high-throughput pipelines** using Star/Snowflake schemas, SCD Type 1/2, and high-performance SQL stored procedures ensuring 99.9% data availability SLA.
- **Implemented automated data quality audit frameworks and ML anomaly detection** with 95% accuracy, mitigating ETL discrepancies and reducing operational troubleshooting turnaround by over 80%.

Graduate Research Data Scientist — **May 2022 – Present**

- **Applied machine learning algorithms and statistical techniques** to complex datasets during graduate studies at the University of Cincinnati, achieving a 3.95/4.0 GPA with a focus on advanced data science.
- **Developed reproducible data science analytics pipelines** leveraging Python and modern statistical frameworks to translate complex analytics output into actionable business recommendations.
- **Conducted exploratory data analysis and predictive modeling** utilizing large-scale datasets, establishing robust methodologies for feature engineering, model validation, and metric visualization.

Process & Data Operations Engineer — **May 2022 – Present**

- **Optimized complex data collection and operational workflows** by applying engineering principles to large-scale industrial datasets and establishing rigorous performance baselines.
- **Collaborated with cross-functional global engineering teams** to gather technical requirements, streamline data processes, and ensure adherence to rigorous quality standards.
- **Synthesized complex analytical findings into executive reports** to drive data-backed decision making and continuous improvement initiatives across operational lifecycles.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.